

Raiya Jessani

COMPUTER SCIENCE

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SKILLS

- **Security:** Penetration Testing, Authentication & Authorization, Firewalls, SIEM, Incident Response
- **Languages:** Python, Java, C, C++, Kotlin, Lua, JavaScript, SQL, Bash/Shell Scripting
- **AWS:** EC2, RDS, EBS, IAM, CloudWatch, S3, Lambda, VPC, CloudFormation, SNS/SQS, Auto Scaling & Load Balancers
- **Cloud:** Virtual machines, Traffic distribution, Network routing, Containerization, Infrastructure as code, Auto provisioning

EXPERIENCE

TD Bank

Jan 2026 – Apr 2026

Security Engineer

Toronto, ON

- Led migration of **application security vulnerability workflows** from unreliable batch Python scripts to **live ServiceNow scanning** across 50+ services, eliminating silent scan failures and ensuring vulnerabilities were caught at the point of code push.
- Triaged 50 daily **security findings**, diagnosing false positives including a Jira character limit overflow silently breaking ticket creation. Resolved with a pipeline code fix, maintaining **clean vulnerability tracking** across 50+ services.
- Led **security risk investigations** across **KRI and DCJ/EPRF** frameworks, improving security metrics accuracy for leadership.

Amazon Web Services (AWS)

May 2025 – Aug 2025

Cloud Engineer

Vancouver, BC

- Built an **RDS/EBS Performance Analyzer** using a **fine-tuned AI model** and an adversarial **two-model debate architecture** trained on historical support cases, cutting 15–20 minutes per case and delivering faster resolutions to AWS customers.
- Explored 30+ **AWS services** via labs and projects, strengthening skills in cloud architecture and scalable integrations.
- Completed a **CIRT (incident response)** rotation on live AWS customer cases, primarily **compromised IAM credentials and account takeovers**, gaining direct exposure to **identity-based attack containment** and remediation at cloud scale.

Hitachi Rail

Sept 2024 – Dec 2024

Software Developer

Toronto, ON

- Diagnosed and resolved a **data corruption issue** in a **C logging system** for rail zone controllers (caused by unbounded queue consumption). Fixed by designing a 100ms event batching class on existing real-time timers, cutting debugging time 40%
- Built log collection and **metadata aggregation pipelines in Python and Shell**, producing event telemetry underpinning **security monitoring, incident response, and forensic analysis**, saving 15+ hours weekly.
- Created FitNesse testing workflows validating log output format and signal ordering, catching 50+ defects pre-deployment

TD Bank

Jan 2023 – May 2023

Data Engineer

Toronto, ON

- Automated **ServiceNow workbook updates** using VBA, cutting processing time 80% across 10K+ records and boosting accuracy.
- Built a **Confluence knowledge** base with 50+ resources, increasing onboarding speed 40%.

TD Bank

May 2022 – Aug 2022

Mobile Software Engineer

Toronto, ON

- Enhanced **Android banking app** with Java/Kotlin, delivering new features and improvements to 8M+ active users.

TECHNICAL PROJECTS

Malware Creation — Security

- Built a self-replicating worm in C++ in a sandboxed lab environment. Studied open-source malware to understand propagation mechanics, process hiding, and OS-level file system interaction, developing the mindset for threat analysis and secure code review.

File Encryption System — Cryptography

- Built a command-line file encryption tool in Python using a hybrid AES and RSA model (AES for bulk encryption, RSA to wrap the key) with fresh key generation per session to limit exposure, tested across multiple file types to validate bit-for-bit decryption integrity.

EDUCATION

University of Waterloo, Bachelor of Computer Science, Honours, Co-op

Dec 2027

Specialization in Business Administration + Specialization in Human-Computer Interaction

CERTIFICATIONS

AWS Certified Security – Specialty – Cloud security best practices and IAM, *40 hours completed*

AWS Certified Generative AI – Specialty – AI services with cloud infrastructure integration, *80 hours completed*

AWS Certified Cloud Practitioner – Foundational AWS knowledge and core services, *30 hours completed*

AWS Certified Solutions Architect – Associate – Designing scalable, cost-efficient architectures, *100 hours completed*

CompTIA Security+ – Cybersecurity fundamentals and risk management, *20 hours completed*